

GIGABYTE GeForce RTX 4060 Ti EAGLE 8G

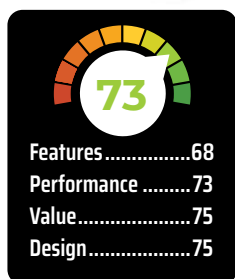
A misstep with the memory

It's no secret that the mid-range segment of graphics cards is something that GPU manufacturers are not fond of. We don't see mid-tier cards for close to a year after the flagships have launched. And even when the mid-tier does launch, there will be some feature set reductions or unnecessary bottlenecks introduced to make the GPU less appealing. That's been the case with both NVIDIA and AMD. Because, once you factor in technologies such as DLSS and FSR, there really is no need to buy a flagship card these days. And the RTX 4060 Ti that we're looking at today, is no different. The RTX 4060 Ti cards are priced around INR 38,000 in India and they're competing with the likes of the RX 6750 XT, 3060 Ti and the 3070, all of which are pretty decent cards.

NVIDIA is targeting 1080p and 1440p gamers with the RTX 4060 Ti and the hardware configuration has been designed in such a manner. We've got two memory variants with the 4060 Ti, of which, we're looking at the 8 GB variant. Either way, the memory interface remains at 128-bit which will undoubtedly limit some of this card's potential. Let's see how it performs.

SPECIFICATIONS

The RTX 4060 Ti, like the rest of the RTX 40 series, is built on NVIDIA's Ada Lovelace architecture, which is the successor to the Ampere architecture used in the previous generation of RTX 30-series cards. At the heart of the 4060 Ti is the AD106 GPU which is built



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using the same TSMC 4N node which is actually a 5 nm node. Compared to its bigger siblings,

the AD106 is a much smaller GPU with a die size of 188 mm² and about 22.9 billion transistors.

All of these transistors are packed into 34 Streaming Multiprocessors, each of which has 128 CUDA cores. So there are a total of 4352 CUDA Cores. A few years back, this would have been the flagship tier but we've seen quite the jump in terms of core counts over these last years. And soon enough, this might be the norm in entry-level graphics cards. Or we might not see entry-level cards altogether considering how integrated graphics on processors are improving by leaps and bounds. The Ada Lovelace introduces several new features and improvements, such as better ray tracing capabilities, faster rendering, and improved power efficiency. The improved power efficiency is due to the switch from Samsung 8nm (RTX 30-series) to the TSMC 4N node

that is about 6 per cent smaller than TSMC's initial 5nm process.

A slightly controversial part of the core configuration of the RTX 4060 Ti is the rather narrow 128-bit memory interface. GDDR6 is much better than the older GDDR5 memory standard in a lot of ways, including data compression. This is one of the reasons why it can achieve good performance even with a slightly narrower bus interface. However, 128-bit is a little too low. It's as if the card has been purposefully bottlenecked so that we may not see better performance at higher resolutions. However, the unfortunate side-effect is that it does lose out on performance even at 1080p simply

because game developers have also become lazy with memory optimisation. Each new video game launching nowadays has high-res textures which take up too much memory. So you need to have a wider memory bus and plenty of memory. Even 8 GB of VRAM is proving to fall short for 1080p gaming with 'ultra' settings. As of now, it's just a few games that need more than 8 GB at 1080p but the day isn't far when most games will need more than 8 GB. And if you factor in the lifecycle of a mid-range card being 3-5 years, this becomes more of a concern.

We've mentioned this before as well. Ada Lovelace gets a massive bump in the L2 cache. The 4060 Ti comes with a sizable 32 MB of L2 cache, which reduces memory access latency and improves overall performance. This is not significantly less than what the RTX 4070 comes with. The added benefit of a larger cache is that a lot of computational workload will not be sent to the much slower GDDR6 memory. DLSS features such as Frame Generation and the upcoming features that are part of the DLSS 3.5 upgrade will benefit a lot because of this.

Lastly, the RTX 4060 Ti like all the other RTX 40-series cards also supports AV1 hardware encoding, a next-generation video codec that offers improved compression efficiency and video quality compared to previous codecs. This